

A Survey of Offline and Preference-Based Reinforcement Learning for Robotics

InteractiveSurvey

Abstract

The rapid advancement of robotic systems in unstructured environments necessitates learning paradigms that ensure safety and efficiency without relying on risky online exploration or explicit reward engineering. Consequently, the field is increasingly converging on hybrid approaches that integrate offline reinforcement learning with preference-based methods to align agent behavior with human intent. This survey provides a comprehensive examination of these synergistic frameworks, specifically focusing on the integration of heterogeneous human feedback ranging from explicit demonstrations to implicit physiological markers. We systematically analyze how combining offline priors with iterative preference refinement overcomes the limitations of distributional shifts and sample inefficiency inherent in isolated methodologies. Key findings highlight strategies for decoding neural signals, modeling bounded human rationality, and employing adaptive curriculum learning to construct robust, dense reward landscapes from sparse interactions. Furthermore, the review details dynamic user modeling techniques that enable personalized policy evolution, facilitating stable convergence in complex, partially observable settings. By synthesizing these diverse modalities, the survey establishes a structured taxonomy that clarifies the relationships between disparate methods for handling noisy feedback and multi-signal fusion. Ultimately, this work serves as a foundational reference for developing scalable, human-aligned robotic systems capable of safe deployment in real-world dynamic environments.

1 Introduction

The rapid advancement of robotic systems in unstructured environments has intensified the need for learning paradigms that can operate safely and efficiently without extensive online interaction.

Traditional reinforcement learning, while powerful, often relies on dense, engineered reward functions and risky trial-and-error exploration that are impractical for real-world physical deployment. Consequently, the field has increasingly turned toward offline reinforcement learning, which leverages pre-collected datasets to initialize policies, and preference-based methods, which align agent behavior with human intent through comparative feedback [1]. These approaches mitigate the dangers of online exploration and the difficulties of explicit reward specification, yet they individually suffer from significant limitations regarding distributional shifts, sample inefficiency, and the ambiguity inherent in sparse human signals.

This survey paper provides a comprehensive examination of the convergence between offline and preference-based reinforcement learning specifically tailored for robotics [1]. We focus on the critical challenge of integrating heterogeneous human feedback-ranging from explicit demonstrations and corrections to implicit physiological markers-into robust learning frameworks. By analyzing how these modalities complement one another, the review highlights strategies that overcome the brittleness of pure imitation learning and the high query complexity of standard preference optimization. The central thesis posits that hybridizing offline priors with iterative preference refinement offers a viable pathway toward deploying adaptable, safe, and personalized robotic agents in complex, dynamic settings where traditional methods fail.

The initial portion of this review investigates the mechanisms for integrating explicit and implicit human signals to construct robust reward landscapes [2]. We explore frameworks that synthesize demonstrations with preference corrections, treating initial trajectories as behavioral priors while utilizing pairwise comparisons to resolve ambiguities and correct suboptimal segments. Further-

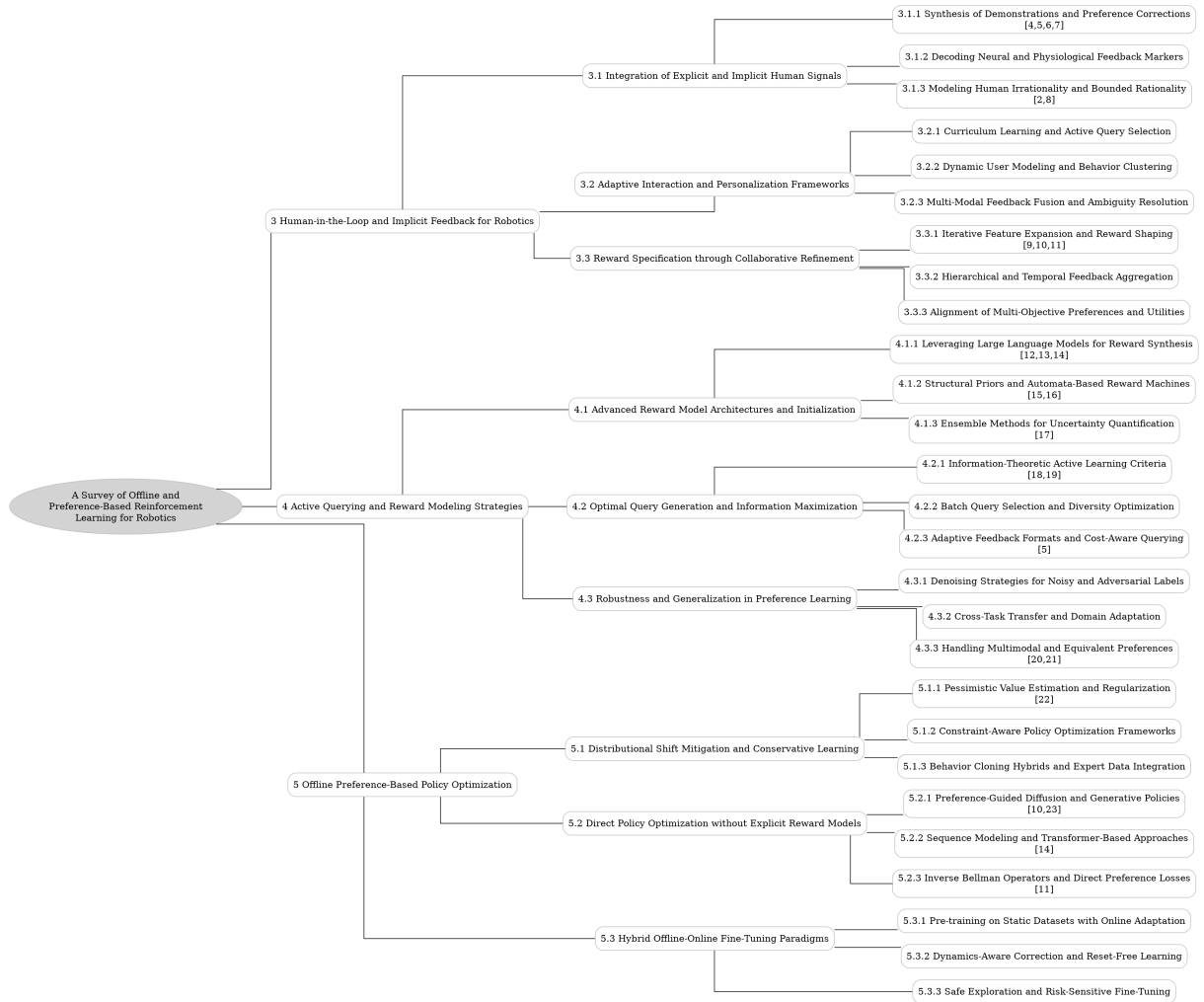


Figure 1: The outline of our survey: A Survey of Offline and Preference-Based Reinforcement Learning for Robotics

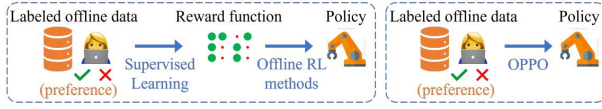


Figure 2: Chart from 'Beyond Reward: Offline Preference-guided Policy Optimization' (arXiv:2305.16217)

more, the discussion extends to decoding neural and physiological feedback markers, such as heart rate variability and cognitive load, translating high-dimensional biological data into quantifiable reward functions. This section also addresses the necessity of modeling human irrationality and bounded rationality, employing thermodynamic frameworks and energy-based models to interpret noisy, satisficing human behaviors as probabilistic expressions of underlying intent rather than erroneous data.

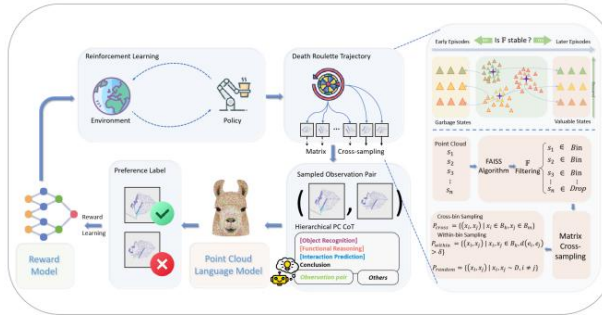


Figure 3: Chart from 'PRISM: Preference Refinement via Implicit Scene Modeling for 3D Vision-Language Preference-Based Reinforcement Learning' (arXiv:2503.10177)

Subsequently, the survey details adaptive interaction and personalization frameworks designed to enhance sample efficiency and generalization. We examine curriculum learning strategies that structure training from simple to complex sub-tasks, coupled with active query selection methods that maximize information gain from limited human interactions. The analysis further covers dynamic user modeling and behavior clustering, which shift from static reward functions to adaptive systems capable of evolving with online interactions and grouping users based on latent behavioral patterns. Additionally, we discuss multi-modal feedback fusion techniques that integrate heterogeneous signals, such as physical interventions and neural indicators, to resolve ambiguities and construct comprehensive reward landscapes that accelerate convergence in stochastic environ-

ments.

The final thematic section focuses on reward specification through collaborative refinement, emphasizing iterative feature expansion and reward shaping. This involves dynamically augmenting state representations with intrinsic signals derived from human feedback to transform sparse extrinsic rewards into dense, informative landscapes. By extending preference queries to cover state-skill transitions over longer horizons and incorporating implicit neural feedback, these methods regularize agent trajectory distributions and stabilize learning in partially observable environments. Together, these diverse methodologies illustrate a cohesive progression toward systems that can continuously refine their understanding of task objectives through nuanced, collaborative human-robot interaction.

The primary contribution of this work lies in its systematic synthesis of recent advancements at the intersection of offline and preference-based learning for robotics [3]. Unlike previous reviews that treat these domains in isolation, this survey uniquely categorizes and analyzes hybrid approaches that leverage the strengths of both paradigms to address real-world deployment challenges. We provide a critical assessment of current techniques for handling noisy human feedback, modeling bounded rationality, and fusing multi-modal signals, offering a structured taxonomy that clarifies the relationships between disparate methods. Ultimately, this paper serves as a foundational reference for researchers aiming to develop more efficient, safe, and human-aligned robotic learning systems, identifying key open problems and guiding future inquiry toward scalable collaborative intelligence.

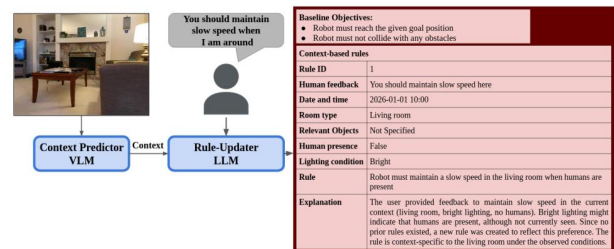


Figure 4: Chart from 'Interpreting Context-Aware Human Preferences for Multi-Objective Robot Navigation' (arXiv:2603.17510)

2 Human-in-the-Loop and Implicit Feedback for Robotics

2.1 Integration of Explicit and Implicit Human Signals

2.1.1 Synthesis of Demonstrations and Preference Corrections

The synthesis of demonstrations and preference corrections addresses the inherent limitations of relying solely on high-quality expert trajectories or sparse comparative feedback. While demonstrations provide a robust initialization for policy learning by establishing a behavioral prior, they often suffer from distributional shifts and an inability to correct suboptimal segments within long-horizon tasks [4]. Conversely, preference-based feedback offers a mechanism for fine-grained reward shaping without requiring explicit reward engineering, yet it can be sample-inefficient when learning from scratch [5]. Contemporary approaches increasingly hybridize these modalities, utilizing demonstrations to anchor the policy near feasible regions of the state space while employing preference queries to iteratively refine the reward function and resolve ambiguities that demonstrations alone cannot address.

$$ACC = (\sum_{\tau_{\hat{w}}} r - \sum_{\tau_w} r)^2$$

Figure 5: Chart from 'INFERRING PREFERENCES FROM DEMONSTRATIONS IN MULTI-OBJECTIVE REINFORCEMENT LEARNING' (arXiv:2409.20258)

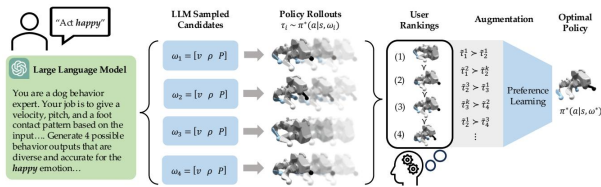


Figure 6: Chart from 'Efficiently Generating Expressive Quadruped Behaviors via Language-Guided Preference Learning' (arXiv:2502.03717)

Advanced frameworks integrate these signals by treating demonstrations as implicit preference data over entire trajectories, thereby reducing the initial uncertainty in the reward landscape before transitioning to denser, state-level or action-level correc-

tions [6]. This sequential strategy leverages the information density of demonstrations during early training phases when the agent's belief over optimal behavior is broad, subsequently shifting to pairwise comparisons to minimize residual uncertainty in specific, ambiguous regions. By localizing suboptimal segments within demonstrated trajectories, methods can generate counterfactual preference pairs that target credit misassignment, ensuring that critic updates are directed toward systematic errors rather than noise. This calibration prevents the propagation of localization errors and suppresses value overestimation more effectively than global trajectory ranking.

$$R(\xi) = \omega \cdot \sum_{s \in \xi} \phi(s) = \omega \cdot \Phi(\xi)$$

Figure 7: Chart from 'Learning Reward Functions from Diverse Sources of Human Feedback: Optimally Integrating Demonstrations and Preferences' (arXiv:2006.14091)

Furthermore, the integration of physical interventions and corrective demonstrations allows for dynamic policy updates that adapt to real-time human feedback during execution [7]. Instead of treating corrections merely as additional replay samples, modern algorithms synthesize them with preference labels to shape the policy gradient directly, balancing exploration with adherence to human intent. This synthesis enables agents to overcome the brittleness of pure imitation learning, which struggles with out-of-distribution states, while mitigating the high query complexity of pure preference-based reinforcement learning. The resulting hybrid paradigms facilitate more natural human-robot collaboration, where coarse behavioral cloning provides a safety baseline, and nuanced preference corrections drive the policy toward optimal, personalized performance in complex manipulation tasks.

2.1.2 Decoding Neural and Physiological Feedback Markers

Decoding neural and physiological feedback markers involves translating high-dimensional biological signals into quantifiable reward functions for robotic learning. Traditional approaches often rely on explicit human corrections, yet recent methodologies prioritize implicit feedback derived from



Figure 8: Chart from 'GAN-Based Interactive Reinforcement Learning from Demonstration and Human Evaluative Feedback' (arXiv:2104.06600)

physiological states such as Heart Rate Variability (HRV) and cognitive load. By analyzing deviations from resting baselines, systems can infer deleterious mental states or physical discomfort without interrupting the interaction flow. This shift enables continuous monitoring of human-robot co-efficiency, where metrics like reaction time and attention distribution serve as proxies for ergonomic alignment and task success.

To process these complex signals, deep neural architectures map raw physiological data into low-dimensional latent spaces suitable for reinforcement learning. Encoder-decoder frameworks, often utilizing multi-layer perceptrons or convolutional layers with ReLU activations, extract feature representations that correlate with human preference. Unlike linear weighted sums of pre-designed features, these non-linear models capture intricate relationships between biomarkers and task statistics. For instance, specific network configurations encode state trajectories into binary classifiers or continuous reward estimates, allowing the agent to distinguish between hazardous and safe operational zones based on the user's subconscious physiological responses.

The integration of these decoded markers significantly enhances sample efficiency in environments where engineering explicit reward functions is prohibitive. By treating physiological stress or facial expressions as dense feedback signals, agents can overcome the sparsity inherent in real-world data collection. Experimental results indicate that methods leveraging implicit biological feedback achieve faster convergence and superior performance compared to baseline algorithms relying solely on explicit demonstrations. Furthermore, adapting to varying noise levels in physiological data ensures robustness, enabling autonomous systems to refine their policies by estimating true expected returns through the continuous interpretation of human behavioral and biological cues.

2.1.3 Modeling Human Irrationality and Bounded Rationality

Modeling human irrationality and bounded rationality is critical for aligning reinforcement learning agents with realistic human feedback, as users frequently deviate from optimal utility maximization due to cognitive constraints and noise. Traditional approaches often assume perfect rationality, yet empirical evidence suggests humans exhibit satisficing behaviors or follow Boltzmann-rational distributions where action probabilities are proportional to exponential reward values. This thermodynamic framework effectively captures the stochastic nature of human decision-making without imposing rigid assumptions, allowing algorithms to interpret sub-optimal demonstrations as probabilistic expressions of underlying preferences rather than erroneous data [8].

$$R(\xi) := \sum_{t=0}^T r(s_t),$$

Figure 9: Chart from 'Learning Preferences for Interactive Autonomy' (arXiv:2210.10899)

To address the computational challenges inherent in complex human models, recent methodologies employ lower-bound optimization and implicit behavior policies to enhance tractability. By utilizing energy-based models to approximate ex-

PLICIT policies, systems can recover robust estimates of human intent even when feedback is sparse or noisy. Furthermore, incorporating parameters for mental and physical effort enables the construction of cost functions that reflect the user’s tendency to minimize exertion. These adjustments allow agents to distinguish between genuine preference signals and artifacts of human fatigue or limited working memory, thereby improving the fidelity of reward shaping in long-horizon tasks [2].

Advanced frameworks also integrate cognitive biases, such as risk aversion and loss sensitivity, through cumulative prospect theory transformations to better simulate human-robot teaming dynamics. By tuning rationality coefficients and noise levels, these models adapt to varying degrees of user consistency, ensuring robustness across different interaction modalities ranging from explicit corrections to implicit neural signals. Ultimately, accurately characterizing the spectrum of human rationality facilitates more efficient preference learning, enabling agents to generalize effectively from limited feedback while maintaining safety and alignment with diverse user expectations in real-world deployments.

2.2 Adaptive Interaction and Personalization Frameworks

2.2.1 Curriculum Learning and Active Query Selection

Curriculum learning in human-in-the-loop reinforcement learning addresses the challenge of sparse rewards by structuring the training process from simple to complex subtasks. By decomposing high-dimensional objectives into manageable skill hierarchies, agents can leverage human feedback to extract structured behavioral priors from noisy offline datasets. This staged approach mitigates the instability associated with naive exploration in physical environments, allowing reward classifiers to effectively guide policy optimization even when dense supervision is unavailable.

Active query selection complements this curriculum by optimizing the information gain derived from limited human interactions. Rather than relying on random or static querying, advanced methods employ uncertainty estimation and information maximization strategies to identify states where human preference labels yield the highest reduction in reward function ambiguity. Techniques such as entropy-guided sample selec-

tion ensure that queries are generated only when the agent’s confidence is low, significantly reducing the cumulative time burden on human operators while maintaining robust convergence rates.

The integration of these paradigms enables sample-efficient frameworks capable of adapting to dynamic real-world constraints. By synchronizing the progression of task difficulty with the strategic acquisition of preference data, systems can achieve high performance with minimal query complexity. Recent advancements demonstrate that classifiers with moderate accuracy, when coupled with active learning mechanisms, suffice to guide effective policy learning, facilitating the transfer of skills from offline pre-training to online fine-tuning without catastrophic degradation.

2.2.2 Dynamic User Modeling and Behavior Clustering

Dynamic user modeling addresses the inherent variability in human preferences by shifting from static reward functions to adaptive frameworks that evolve with online interactions. Traditional approaches often struggle with sparse signals in physical environments, necessitating methods that leverage real-time intervention data rather than relying solely on offline trajectory comparisons. By integrating online gate updates and preference-guided actor refinements, these systems regularize policy learning against purely online states, ensuring that the model captures shifting user intents without requiring exhaustive pre-collected datasets.

Behavior clustering further enhances sample efficiency by grouping users or tasks based on latent behavioral patterns, allowing for the transfer of learned priors across similar contexts. Techniques such as two-stage clustering analyze action prediction sequences to identify distinct user types, enabling the system to apply specific behavioral models rather than a single generalized policy. This stratification is critical in collaborative settings where different operators exhibit unique strategies, as it permits the algorithm to localize suboptimal segments and tailor feedback queries to the specific cluster characteristics, thereby reducing the overall burden on human annotators.

The synergy between dynamic modeling and clustering facilitates robust generalization from offline demonstrations to online deployment, mitigating the risk of catastrophic degradation when encountering novel states. By employing

lightweight progress estimators trained in a self-supervised manner, agents can effectively distinguish between task-specific nuances and universal objectives. Although subject-wise variability remains a challenge, combining implicit feedback decoding with active querying strategies ensures that classifiers maintain moderate yet consistent accuracy, ultimately guiding effective policy improvement even under limited human feedback constraints.

2.2.3 Multi-Modal Feedback Fusion and Ambiguity Resolution

Multi-modal feedback fusion integrates heterogeneous signals, ranging from explicit physical corrections via SpaceMouse interventions to implicit neural indicators like EEG patterns, into a unified learning framework. By encoding ego-centric and external visual streams alongside proprioceptive states through shared latent spaces, systems can correlate sparse binary preferences with dense trajectory modifications. This synthesis allows the agent to leverage the high information density of physical overrides while simultaneously interpreting subtle affective states, thereby constructing a more comprehensive reward landscape than single-modality approaches permit.

Ambiguity resolution within this fused landscape addresses the inherent noise and conflicting gradients present in human-in-the-loop scenarios. When explicit corrections contradict implicit neural signals or when demonstration data exhibits multi-modal distributions, the system employs probabilistic modeling to estimate the true intent behind the feedback. Techniques such as counterfactual corrections and uncertainty-aware decoding help distinguish between genuine user dissatisfaction and sensor noise, effectively narrowing the feasible region of optimal policies. This process often involves minimizing temporal-difference errors against preference-calibrated targets that weigh feedback reliability based on historical consistency and signal confidence.

The computational feasibility of resolving these ambiguities relies on reformulating complex bilevel optimization problems to avoid expensive second-order derivatives. By utilizing first-order approximations and parallelizable objective functions, modern frameworks enable real-time updates to both actor and critic networks without incurring prohibitive latency. This efficiency is critical for deploying robust fusion mechanisms in dy-

namic environments, ensuring that the robot can adapt its behavior instantaneously to resolve conflicting inputs. Consequently, the integration of diverse feedback modalities not only accelerates convergence but also enhances the robustness of policy learning against the stochastic nature of human interaction.

2.3 Reward Specification through Collaborative Refinement

2.3.1 Iterative Feature Expansion and Reward Shaping

Iterative feature expansion addresses the inherent sparsity of extrinsic rewards in robotic learning by dynamically augmenting the state representation with intrinsic signals derived from human feedback [9]. Rather than relying on static, pre-defined reward functions, this approach progressively refines the feature space to capture nuanced user preferences that are initially latent. By treating the reward function as a linear combination of evolving features, the system can adaptively weight task-agnostic dimensions, effectively transforming sparse success signals into a dense, informative landscape that guides policy optimization toward desired behaviors without exhaustive exploration.

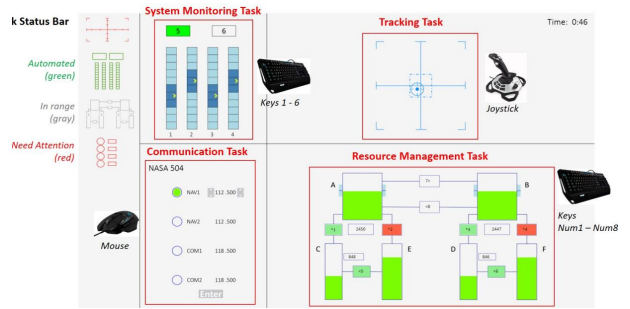


Figure 10: Chart from 'Offline Reinforcement Learning Framework for State Space and Reward Function Selection in Human-Robot Teaming' (arXiv:2602.02530)

Reward shaping within this framework leverages preference-based learning to construct auxiliary objectives that regularize the agent's trajectory distribution [10]. Instead of operating solely on atomic state-action transitions, modern methodologies extend preference queries to cover state-skill transitions over longer time horizons, thereby stabilizing the learning process in partially observable environments. The integration of implicit neural feedback, such as error-related potentials, fur-

ther accelerates convergence by providing continuous evaluative signals that shape the loss function away from undesired regions and toward high-reward modes identified through human interaction.

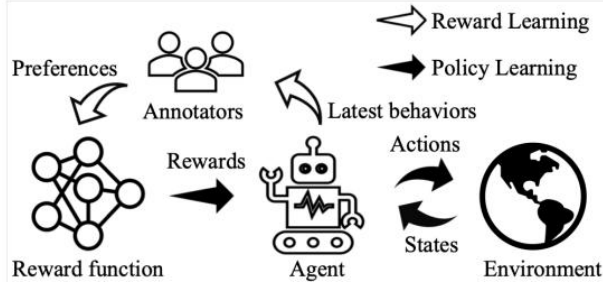


Figure 11: Chart from 'Online Policy Learning from Offline Preferences' (arXiv:2403.10160)

The synergy between iterative expansion and shaping is realized through a cyclic optimization process where the reward model and policy are co-adapted. In each iteration, the system updates the behavioral prior and reward parameters by minimizing a composite loss that balances maximum entropy objectives with human preference alignment [11]. This mechanism ensures that the learned reward function not only reflects the user's latent intent but also maintains sufficient exploration coverage, ultimately enabling sample-efficient acquisition of complex skills that would be intractable under standard sparse-reward regimes.

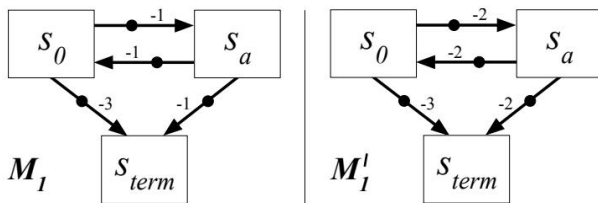


Figure 12: Chart from 'Models of human preference for learning reward functions' (arXiv:2206.02231)

2.3.2 Hierarchical and Temporal Feedback Aggregation

Hierarchical and temporal feedback aggregation addresses the challenge of integrating sparse, delayed human signals into dense reward structures for complex manipulation tasks. Rather than treating trajectories as monolithic units, advanced frameworks decompose intervention-containing episodes into distinct segments: normal execution, suboptimal deviations, and human-corrected

intervals. This segmentation allows the system to isolate specific failure modes and apply targeted credit assignment, ensuring that positive reinforcement is attributed only to the corrected sub-trajectories while penalizing preceding errors. By converting heterogeneous interaction logs into structured state-skill sequences, agents can learn high-level policies that effectively bridge the gap between atomic actions and long-horizon goals.

Temporal aggregation mechanisms further refine this process by aligning feedback with the appropriate time scales of task execution. In settings where humans provide binary success labels or pairwise preferences over extended horizons, the system must propagate these signals backward through time to update value estimates for earlier states. Methods employing high update-to-data ratios utilize asynchronous actor-critic architectures to continuously integrate these temporal preferences, often leveraging pretrained encoders to extract robust state representations from raw observations. This approach mitigates the sparsity of terminal rewards by generating intermediate dense rewards based on the likelihood of segment success, thereby accelerating convergence in environments where direct supervision is infrequent.

The synthesis of hierarchical decomposition and temporal propagation enables robust learning from limited human intervention. By aggregating feedback across multiple levels of abstraction, from low-level motor corrections to high-level task completion indicators, agents achieve greater sample efficiency and generalization. This multi-scale integration ensures that the learned policy respects both the immediate physical constraints identified during interventions and the overarching logical structure of the task. Consequently, the agent develops a nuanced understanding of task dynamics, capable of recovering from suboptimal states and executing complex, multi-stage procedures with minimal additional human guidance.

2.3.3 Alignment of Multi-Objective Preferences and Utilities

The alignment of multi-objective preferences necessitates moving beyond scalar reward maximization toward frameworks that accommodate conflicting stakeholder utilities. Traditional approaches often rely on pairwise trajectory comparisons to infer a latent reward function, yet these methods struggle when preferences are non-stationary or when objectives are incommensu-

rable. Recent advancements address this by formulating alignment as a bilevel optimization problem, where the outer loop learns a reward model consistent with human feedback while the inner loop optimizes the policy. This structure prevents the disentanglement issues seen in earlier works, ensuring that the learned reward function remains robust to policy shifts and accurately reflects the nuanced trade-offs inherent in complex, multi-step tasks.

To handle the complexity of compositional tasks, contemporary methods increasingly specify preferences over skills or trajectory segments rather than entire episodes. This granularity allows for more efficient sample complexity and facilitates the transfer of preferences across domains where feature overlaps exist. By constructing zero-cost preference pairs from human corrective actions and policy deviations, algorithms can directionally penalize Bellman targets for suboptimal segments. This mechanism suppresses Q-value overestimation on the critic side while simultaneously guiding the actor toward behaviors that satisfy specific utility constraints, effectively balancing exploration with strict adherence to safety-critical or ethical requirements defined a priori.

Furthermore, integrating regularization terms derived from bilevel optimization literature enables the system to weigh agent performance against human preference intensity dynamically. Such regularization ensures that the learned reward model does not diverge significantly from previous iterations, maintaining stability during online policy improvement. In interactive settings, this approach proves superior to static offline datasets, as it exploits coarse human interventions without requiring them to match optimal corrective actions exactly. Ultimately, achieving robust alignment requires designing objectives that explicitly account for the dependence of the policy on the reward model, thereby mitigating causal confusion and ensuring that the resulting behavior aggregates multiple utilities in a manner consistent with diverse human values.

3 Active Querying and Reward Modeling Strategies

3.1 Advanced Reward Model Architectures and Initialization

3.1.1 Leveraging Large Language Models for Reward Synthesis

Recent advancements have positioned Large Language Models (LLMs) as pivotal tools for automating reward function synthesis, effectively mitigating the bottlenecks of manual reward engineering and expensive human preference labeling [12]. By leveraging the extensive world knowledge and reasoning capabilities inherent in pre-trained LLMs, researchers can now generate executable reward code or dense reward signals directly from natural language instructions [13]. This paradigm shift enables agents to interpret high-level semantic goals and translate them into precise, task-specific reward structures without requiring iterative human-in-the-loop feedback for every nuance of the desired behavior.

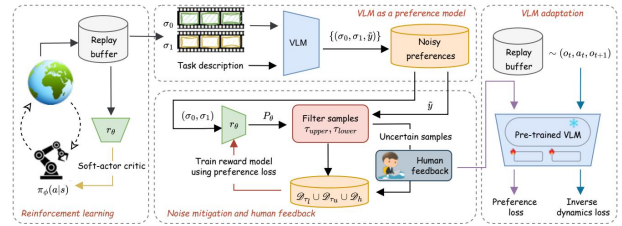


Figure 13: Chart from 'Preference VLM: Leveraging VLMs for Scalable Preference-Based Reinforcement Learning' (arXiv:2502.01616)

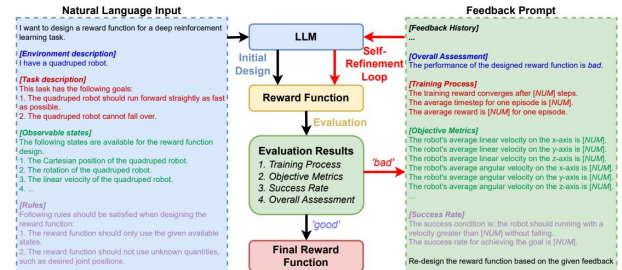


Figure 14: Chart from 'Self-Refined Large Language Model as Automated Reward Function Designer for Deep Reinforcement Learning in Robotics' (arXiv:2309.06687)

A prominent approach within this domain involves utilizing LLMs to iteratively refine reward functions through code generation and self-correction mechanisms. In this framework, the LLM acts as an autonomous reward designer that

proposes candidate reward functions, evaluates agent performance against these functions, and subsequently modifies the code based on observed failures or suboptimal trajectories [13]. This process often incorporates feedback loops where the model analyzes trajectory data to identify discrepancies between intended and actual behaviors, thereby synthesizing increasingly accurate reward models that capture complex temporal dependencies and multi-objective constraints that are difficult to specify manually.

Furthermore, the integration of LLMs facilitates the distillation of human preferences into scalable reward models by generating diverse synthetic preference pairs or pseudo-labels for unlabeled trajectories [14]. This capability significantly enhances sample efficiency in preference-based reinforcement learning by expanding the training distribution beyond limited human queries. By combining LLM-generated reward priors with active learning strategies, systems can focus human annotation efforts on the most ambiguous or informative state-action segments, accelerating policy convergence while ensuring the final reward function remains robustly aligned with human values and intent across diverse environments.

3.1.2 Structural Priors and Automata-Based Reward Machines

Structural priors serve as a critical mechanism for mitigating the sample inefficiency inherent in preference-based reinforcement learning by constraining the hypothesis space of reward functions. Rather than learning rewards solely from scratch via human feedback, these methods integrate domain knowledge or symbolic representations to guide the inference process. By embedding logical constraints or temporal dependencies directly into the reward model architecture, agents can generalize more effectively from sparse preference queries. This approach significantly reduces the number of interactions required to converge on an accurate reward function, particularly in environments where the underlying objective exhibits complex, non-Markovian characteristics that standard scalar reward models struggle to capture.

Automata-based reward machines extend this paradigm by utilizing finite state automata, such as Deterministic Finite Automata (DFA), to formally specify high-level task structures. In this framework, the reward function is not a static mapping from states to scalars but a dynamic sys-

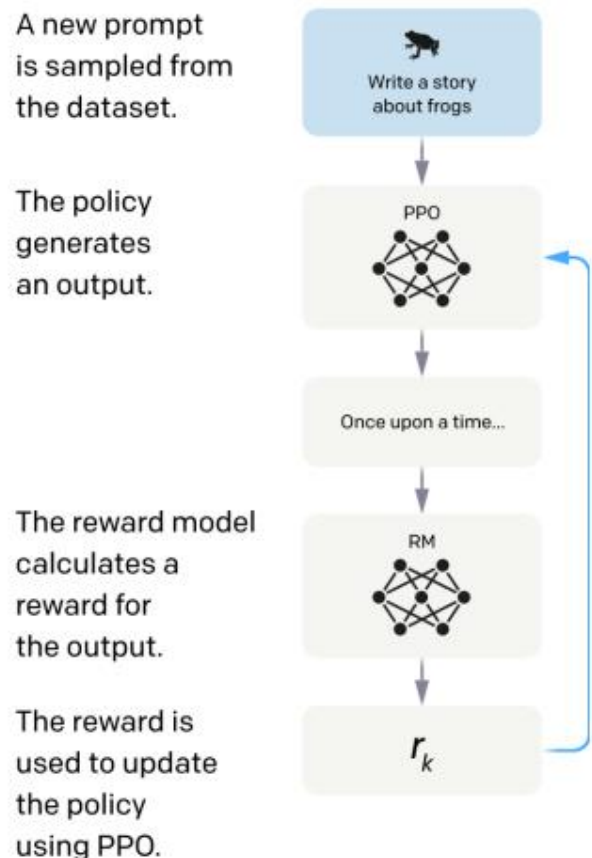


Figure 15: Chart from 'Reinforcement Learning in the Era of LLMs: What is Essential? What is needed? An RL Perspective on RLHF, Prompting, and Beyond.' (arXiv:2310.06147)

tem where rewards are contingent upon transitions between automaton states [15]. This allows for the precise encoding of temporal logic specifications and sequential sub-goals without manual engineering of dense reward signals. The automaton acts as a memory module, tracking progress through abstract task phases and providing structured feedback that aligns agent behavior with long-term objectives, thereby overcoming the limitations of Markovian assumptions in complex decision-making scenarios.

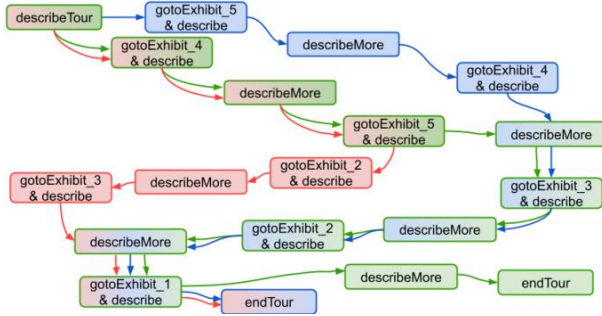


Figure 16: Chart from 'Learning on the Job: Long-Term Behavioural Adaptation in Human-Robot Interactions' (arXiv:2203.10518)

Recent advancements have begun to hybridize these structural approaches with active preference learning, leveraging automata to generate informative trajectory queries rather than defining rewards explicitly [16]. Instead of relying on hand-crafted reward machines, the structural topology of the automaton is used to synthesize diverse trajectory pairs that maximize information gain regarding the human's true intent. This synergy enables the agent to learn non-Markovian reward functions that are both consistent with human preferences and compliant with formal task specifications. Consequently, this integration offers a robust solution for learning in sparse-reward environments, balancing the flexibility of data-driven preference learning with the interpretability and efficiency of symbolic structural priors.

3.1.3 Ensemble Methods for Uncertainty Quantification

Ensemble methods constitute a primary strategy for quantifying epistemic uncertainty in deep reward modeling by leveraging the disagreement among multiple independently trained predictors. Typically, an ensemble of M reward networks is initialized with diverse random seeds or data shuffling to ensure variation in their convergence

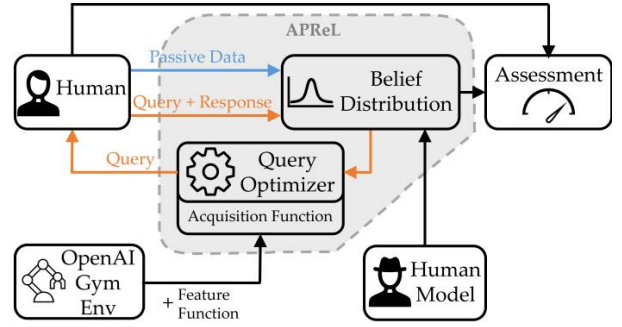


Figure 17: Chart from 'APReL: A Library for Active Preference-based Reward Learning Algorithms' (arXiv:2108.07259)

trajectories. The uncertainty associated with a specific trajectory segment or state-action pair is then estimated via the statistical variance or entropy of the predictions across the ensemble members. High disagreement indicates regions where the model lacks sufficient training data, effectively mapping the decision boundary of the preference function without requiring explicit Bayesian inference.

In the context of active preference learning, this uncertainty estimation drives query selection mechanisms designed to maximize information gain [17]. Rather than selecting segment pairs uniformly, algorithms prioritize comparisons where the ensemble exhibits maximum entropy or disagreement, corresponding to samples nearest the decision boundary where the probability of preference approaches 0.5. This targeted sampling ensures that human feedback is solicited for the most ambiguous cases, theoretically accelerating the reduction of uncertainty in the unlabelled behavior space. Empirical studies suggest that such uncertainty-based sampling schemes significantly outperform naive uniform sampling in complex domains by sharpening decision boundaries more efficiently with fewer queries.

$$c^*(w') = c(P', w').$$

Figure 18: Chart from 'Active Preference Learning using Maximum Regret' (arXiv:2005.04067)

However, relying solely on ensemble disagreement presents challenges, particularly when high uncertainty stems from noisy or meaningless motion segments rather than genuine ambiguity in optimal behavior. In such scenarios, selecting

queries based purely on variance may lead to inefficient learning or the incorporation of incorrect labels if the compared segments lack distinct semantic differences. Consequently, advanced frameworks often augment pure uncertainty metrics with diversity constraints or motion information analysis to ensure selected queries are both informative and meaningful. While ensemble methods offer robust empirical performance and ease of implementation compared to Bayesian dropout, they incur higher computational costs due to the need to maintain and update multiple models simultaneously.

3.2 Optimal Query Generation and Information Maximization

3.2.1 Information-Theoretic Active Learning Criteria

Information-theoretic active learning criteria prioritize query selection by maximizing the expected reduction in uncertainty regarding the underlying reward function parameters. Central to this approach is the formulation of information gain, typically defined as the difference between the prior and posterior entropy of the reward distribution. By leveraging Bayesian frameworks, these methods estimate the posterior probability over reward weights given historical preference data, allowing the agent to identify trajectory pairs that yield the highest mutual information between the potential human feedback and the latent reward model [18].

A predominant strategy within this domain involves maximizing the expected information gain to shrink the volume of the feasible weight space, often approximated through ensemble methods or Laplace approximations to handle computational intractability. Unlike simple uncertainty sampling, which may select ambiguous queries near decision boundaries without guaranteeing significant model updates, information-theoretic objectives explicitly target queries that induce the largest shift in the belief distribution. This ensures that each human interaction provides maximal discriminative power, effectively distinguishing between competing reward hypotheses and accelerating convergence toward the true preference structure [19].

Despite their theoretical robustness, these criteria face challenges related to computational complexity, particularly when evaluating integrals over high-dimensional posterior distributions. Conse-

$$\pi_0 = \arg \max_{\pi} \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^T \gamma^t r_{\phi}(s_t, a_t) \right]$$

Figure 19: Chart from 'PBŠ: Preference Space Exploration via Population-Based Methods in Preference-Based Reinforcement Learning' (arXiv:2506.13741)

quently, recent advancements have introduced approximations such as using the variance of ensemble predictions or employing greedy optimization strategies to select batch queries that balance informativeness with diversity. While greedy selection lacks strict theoretical optimality guarantees in pathological settings, empirical evidence suggests it offers a practical trade-off, enabling efficient online reward learning in complex robotic environments where real-time query generation is essential.

3.2.2 Batch Query Selection and Diversity Optimization

Batch query selection in preference-based reinforcement learning aims to construct informative subsets of trajectory pairs that maximize learning efficiency while minimizing human labeling effort. Traditional approaches often rely on uncertainty metrics, such as ensemble disagreement or predictive entropy, to identify samples where the reward model is most ambiguous. However, selecting queries based solely on uncertainty can lead to the presentation of indistinguishable or noisy segments, potentially confusing human annotators and degrading reward model accuracy. Consequently, effective batch selection must balance informativeness with the discriminability of the presented options, ensuring that selected pairs offer clear preference signals rather than merely reflecting model confusion.

To address the challenge of redundancy within selected batches, diversity optimization techniques are employed to ensure broad coverage of the state-action space. Methods such as Medoid Selection leverage clustering algorithms on feature representations to pick the most dissimilar queries, thereby preventing the batch from being dominated by similar trajectories that provide diminishing returns. Furthermore, volume removal metrics and determinantal point processes have been adapted to model the marginal gain of adding specific queries to a batch, explicitly penalizing re-

dundancy. These strategies often incorporate trajectory diversification by sampling from multiple policies or maintaining curiosity buffers, which encourages the inclusion of behaviors that are both unfamiliar to the agent and favored by human evaluators.

Advanced frameworks integrate these principles by jointly optimizing for preference discriminability and trajectory diversity during the batch generation process. By training discriminators to estimate the likelihood of human distinguishability, systems can probabilistically sample queries that are not only uncertain but also semantically distinct enough for reliable labeling. This dual objective mitigates the policy bias inherent in single-policy exploration and ensures that the resulting preference data covers a wider mode of the reward distribution. Ultimately, synthesizing uncertainty estimation with geometric diversity constraints allows for the construction of compact, high-value query batches that accelerate convergence and improve the robustness of the learned reward functions.

3.2.3 Adaptive Feedback Formats and Cost-Aware Querying

Adaptive feedback formats move beyond static pairwise comparisons to dynamically select query structures that maximize information gain while minimizing human cognitive load. By leveraging uncertainty estimates from reward ensembles or Gaussian processes, systems can identify regions of the state space where preference ambiguity is highest, prompting targeted queries such as triplet rankings or scalar scores instead of binary choices. This adaptivity ensures that each interaction yields maximal discriminative power, effectively reducing the total number of queries required to converge on an accurate reward model.

Cost-aware querying strategies further refine this process by explicitly incorporating the economic and temporal costs of human annotation into the selection criterion. Rather than indiscriminately sampling high-uncertainty segments, these methods evaluate a utility function that balances expected information gain against the anticipated effort required for labeling. Techniques such as threshold-based skipping allow the system to bypass queries with negligible reward differences, preventing annotator fatigue and avoiding unreliable data generated from indistinguishable trajectory pairs, thereby enhancing the overall signal-to-

noise ratio of the collected feedback.

Integrating these approaches enables a closed-loop optimization where the query policy evolves alongside the reward learner [5]. Advanced frameworks employ active inference or Bayesian optimization to jointly determine the optimal feedback format and timing, adapting to individual user performance and consistency metrics in real-time. This synergy between adaptive formatting and cost sensitivity not only accelerates convergence in complex, high-dimensional tasks but also makes preference-based reinforcement learning scalable for real-world applications where human feedback is a scarce and expensive resource.

3.3 Robustness and Generalization in Preference Learning

3.3.1 Denoising Strategies for Noisy and Adversarial Labels

Denoising strategies for noisy and adversarial labels primarily focus on mitigating the detrimental impact of incorrect human feedback and inconsistent preference data on policy optimization. A prevalent approach involves employing robust loss functions designed to reduce the influence of outliers, thereby preventing the learning process from being misled by erroneous signals. Techniques such as filtering inconsistent responses or utilizing ensemble methods to aggregate multiple predictions have proven effective in stabilizing training dynamics. By calculating the mode of sampled preference labels or averaging rewards from models with the lowest validation losses, these methods significantly enhance the reliability of policy updates even when underlying data sources are compromised by high levels of noise.

In the context of adversarial imitation learning, specific mechanisms address label corruption by integrating uncertainty estimation directly into the reward modeling framework. Methods often leverage deep ensembles or Bayesian approximations to quantify predictive variance, allowing the system to identify and down-weight uncertain samples located near decision boundaries. This uncertainty-unified framework enables the detection of local maxima in reward landscapes that baseline methods might miss due to noise-induced smoothing. Furthermore, incorporating regularization terms or adopting doubly robust estimators helps balance the trade-off between bias and variance, ensuring consistent performance even when

either the reward model or importance weights are partially inaccurate.

Advanced denoising protocols also extend to active learning scenarios where query efficiency is paramount under noisy conditions. By synthesizing queries that maximize information gain while accounting for potential label corruption, algorithms can achieve near-optimal convergence rates despite adversarial interference. Strategies include injecting controlled noise into policies to generate ranked trajectories for reward extrapolation, which facilitates better-than-demonstrator performance without requiring pristine data. Ultimately, the integration of these noise-robust techniques with off-policy learning frameworks presents a promising direction for developing resilient agents capable of operating effectively in real-world environments characterized by unreliable feedback and adversarial perturbations.

3.3.2 Cross-Task Transfer and Domain Adaptation

Cross-task transfer in reinforcement learning increasingly leverages unsupervised pre-training to extract diverse behavioral priors, significantly enhancing sample efficiency for downstream objectives. By maximizing state coverage without extrinsic rewards, agents develop robust foundational skills that facilitate rapid adaptation to novel tasks. This paradigm shifts the learning burden from scratch training to fine-tuning, where previously acquired representations allow agents to generalize across complex locomotion and manipulation domains with minimal additional data.

Domain adaptation further benefits from hybrid experience mechanisms that integrate expert demonstrations with preference-based feedback to mitigate covariate shift. Architectures employing hybrid replay buffers store both expert trajectories and human-preferred states, ensuring policy updates prioritize high-value segments while maintaining exploration diversity. This approach effectively addresses the scarcity of labeled data by combining off-policy learning with active query selection, allowing agents to navigate dynamic environments where reward structures may vary but logical task constraints remain consistent.

Recent advancements introduce preference-discriminable frameworks that retrain policies from scratch upon reward model updates to eliminate inherited policy bias. By generating trajectory pairs with high discriminability, these methods en-

sure that human feedback provides meaningful signals even when transferring across embodiments with differing dynamics. Consequently, the synergy between modular policy structures and adaptive reward learning enables robust generalization, allowing agents to maintain performance stability despite significant variations in environmental parameters or sensor modalities.

3.3.3 Handling Multimodal and Equivalent Preferences

Handling multimodal preferences requires moving beyond the traditional assumption of a unimodal reward function, as human intent often encompasses diverse, equally valid behavioral modes. Standard pairwise comparison frameworks frequently struggle to capture this complexity, potentially converging to an average behavior that satisfies no specific mode. Recent advancements address this by leveraging ranking-based feedback rather than binary choices, enabling the learning of structured multimodal reward landscapes [20]. By estimating the probability distribution over modes based on comparative feedback, algorithms can distinguish between distinct optimal trajectories instead of collapsing them into a single suboptimal policy.

$$p(\omega \mid \mathcal{R}) \propto p(\omega) \prod_{i=0}^N p(\xi_i \mid Q_i)$$

Figure 20: Chart from 'Improving User Experience in Preference-Based Optimization of Reward Functions for Assistive Robots' (arXiv:2411.11182)

Equally critical is the treatment of equivalent preferences, where human raters perceive two trajectory segments as indistinguishable or equally desirable. Ignoring these equal judgments discards valuable information regarding the flatness of the reward landscape and introduces noise into the learning signal. Novel multi-type preference learning methods explicitly incorporate equal preference labels alongside strict rankings, allowing the model to recognize regions of indifference [21]. This approach enhances sample efficiency by preventing the agent from unnecessarily optimizing within plateaus of the reward function, thereby focusing exploration on states that yield meaningful discriminative feedback.

$$P_\psi[\sigma^0 \succ \sigma^1] = \frac{\exp \sum_t \hat{r}_\psi(s_t^0, a_t^0)}{\sum_{i \in \{0,1\}} \exp \sum_t \hat{r}_\psi(s_t^i, a_t^i)}.$$

Figure 21: Chart from ‘Multi-Type Preference Learning: Empowering Preference-Based Reinforcement Learning with Equal Preferences’ (arXiv:2409.07268)

Integrating these concepts involves unifying human uncertainty with robot uncertainty to robustly handle ambiguous feedback scenarios. Advanced frameworks utilize mixture models or latent variable approaches to represent multiple reward hypotheses simultaneously, updating belief distributions as new multimodal or equivalent data arrives. This synergy allows agents to maintain diverse policy candidates until sufficient evidence resolves the ambiguity. Consequently, robots can adapt to complex human values that vary across contexts or individuals, ensuring that the learned behavior aligns with the full spectrum of acceptable human preferences rather than a narrow, potentially misaligned optimum.

4 Offline Preference-Based Policy Optimization

4.1 Distributional Shift Mitigation and Conservative Learning

4.1.1 Pessimistic Value Estimation and Regularization

Pessimistic value estimation addresses the critical challenge of distributional shift and overestimation bias inherent in offline reinforcement learning from human feedback. Unlike standard point-estimate approaches that rely on gradient ascent to update reward parameters, pessimistic methods explicitly account for uncertainty by constructing confidence sets around the true reward function. This framework posits that when data coverage is partial or human demonstrations are suboptimal, the learning agent should optimize policies against the worst-case reward within a plausible uncertainty set. By penalizing values associated with out-of-distribution actions or states, these methods prevent the agent from exploiting modeling errors, thereby ensuring that the estimated value function serves as a rigorous lower bound on the true expected return.

Regularization techniques serve as the primary mechanism for implementing this pessimism without solving computationally intractable robust op-

timization problems directly. Approaches such as Conservative Q-Learning introduce explicit regularizers to the Q-function objective, effectively lowering Q-values for actions absent from the dataset while preserving values for observed behaviors [22]. Alternative strategies employ parameter-space regularization, including L1 and L2 norms or Bayesian dropout, to constrain the complexity of the reward model and stabilize training dynamics. These regularizers implicitly define an uncertainty penalty, discouraging the policy from deviating significantly from the behavior policy and mitigating the risk of learning spurious correlations from noisy or erroneous human interventions.

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right]$$

Figure 22: Chart from ‘DiSA-IQL: Offline Reinforcement Learning for Robust Soft Robot Control under Distribution Shifts’ (arXiv:2510.00358)

The theoretical justification for these methods rests on proving that the resulting value estimates remain pessimistic with high probability, even under general function approximation. By integrating uncertainty bonuses or modifying the Bellman backup operator to include a penalty term derived from the variance of ensemble predictions or dropout masks, algorithms can achieve robust convergence guarantees. This ensures that the learned policy maximizes performance only where the data provides sufficient support, effectively balancing the trade-off between exploration and safety. Consequently, pessimistic regularization transforms offline learning from human feedback into a stable procedure capable of handling the inevitable imperfections and inconsistencies found in real-world human demonstration datasets.

4.1.2 Constraint-Aware Policy Optimization Frameworks

Constraint-aware policy optimization frameworks address the critical challenge of ensuring safety and feasibility during offline reinforcement learning by explicitly restricting the policy search space. These methods typically employ policy-constraint mechanisms, such as behavior cloning regularization or Hellinger distance bounds, to confine the learned policy within a confidence set derived

from the offline dataset. By limiting exploration to state-action pairs with sufficient coverage, these frameworks prevent the agent from venturing into out-of-distribution regions where value estimates are unreliable, thereby mitigating the risk of catastrophic failures in real-world robotic deployments.

Recent advancements integrate preference-based learning signals to refine these constraints, balancing optimality with strict adherence to safety requirements. Algorithms like StROL modify standard gradient-based learning rules to expand the basins of attraction, ensuring convergence to human true preferences even under suboptimal feedback conditions. Furthermore, theoretical analyses establish partial coverage guarantees by introducing concentrability coefficients and duality gap bounds, which modulate the trade-off between maximizing accumulated rewards and satisfying feasibility constraints defined by linear programming or conservative Q-function regularization.

Practical implementations often leverage model-based approaches or ensemble disagreement to identify and avoid undesirable constraint regions, such as physical obstacles in navigation tasks. By utilizing auxiliary rewards derived from counterfactual predictions or incorporating obstacle avoidance costs into model-predictive control planners, these frameworks achieve robust performance across diverse morphological parameters and environments. Consequently, constraint-aware optimization enables the deployment of policies that not only maximize task success rates but also maintain high levels of trajectory safety and generalization in complex, dynamic settings.

4.1.3 Behavior Cloning Hybrids and Expert Data Integration

Behavior Cloning (BC) hybrids represent a critical evolution in offline learning, addressing the fundamental limitations of pure supervised imitation, such as compounding errors and sensitivity to distributional shift. By integrating BC objectives with reinforcement learning mechanisms, these architectures leverage the stability of expert data while mitigating the risks associated with out-of-distribution actions. Recent approaches often employ pessimistic regularization or advantage-weighted updates to ensure the learned policy remains within the support of the behavior policy, effectively bridging the gap between static demonstration datasets and dynamic decision-making re-

quirements without necessitating active environment interaction.

The integration of expert data within these hybrid frameworks frequently involves sophisticated occupancy matching or adversarial techniques to minimize statistical divergence between learner and expert distributions. Unlike standard BC, which strictly clones actions, hybrid models utilize expert trajectories to shape reward landscapes or constrain policy search spaces, enabling robustness against noisy or sub-optimal demonstrations. This is particularly evident in methods that combine medium-quality replay buffers with expert samples, where the algorithm dynamically balances imitation fidelity with exploratory potential. Such strategies allow agents to surpass the performance ceiling of the behavior policy by extrapolating from high-quality segments while maintaining safety through conservative estimates of unseen state-action pairs.

Furthermore, advanced hybrid systems incorporate hierarchical structures or preference-based feedback to refine policy execution beyond simple action cloning. By conditioning low-level policies on predicted subgoals or utilizing human preference labels to re-weight offline trajectories, these models achieve superior generalization in long-horizon tasks where standard BC fails. The synergy between offline dataset coverage and implicit reward modeling enables agents to navigate complex environments with limited data diversity. Consequently, behavior cloning hybrids have emerged as a dominant paradigm for deploying robotic systems in scenarios where online exploration is prohibitive, offering a scalable pathway to transferable and socially compliant autonomous behaviors.

4.2 Direct Policy Optimization without Explicit Reward Models

4.2.1 Preference-Guided Diffusion and Generative Policies

Preference-guided diffusion models have emerged as a powerful mechanism for synthesizing high-quality trajectories that align with latent human reward structures. By leveraging the strong generative capabilities of diffusion processes, these methods sample candidate behaviors conditioned on a prior distribution over reward parameters, effectively expanding the basin of attraction for convergence toward true human preferences. This

approach modifies the underlying dynamics to rapidly identify state-action pairs that maximize the probability of preferred outcomes, even when explicit reward functions are unavailable or sparse [10].

The integration of generative policies within this framework facilitates robust offline policy optimization by decoupling trajectory generation from environment interaction. Instead of relying on costly online rollouts, algorithms utilize learned transition models to simulate diverse trajectory pairs, which are then evaluated against a Bradley-Terry preference model to quantify uncertainty. By selecting policies that maximize preference uncertainty or align closely with sampled optimal actions, the system iteratively refines its estimate of the reward landscape, ensuring that the learned policy remains consistent with the underlying ground-truth preferences despite limited feedback [23].

$$Q^{(i)}(x_k, u_k) = \tilde{U}(x_k, u_k) + \alpha_0 \mathcal{V}(x_k) + \sum_{j=1}^{\infty} [\tilde{U}(x_{k+j}, h^{(i)}(x_{k+j})) + \alpha_i \mathcal{V}(x_{k+j})].$$

Figure 23: Chart from ‘Reinforcement Learning Control of Robotic Knee with Human in the Loop by Flexible Policy Iteration’ (arXiv:2006.09008)

Furthermore, recent advancements incorporate pessimistic estimation and distribution correction to mitigate exploration errors inherent in offline settings. These techniques regularize the learned policy to stay proximal to the behavior distribution while optimizing for expected utility derived from the generative model. By treating the preference model as a latent utility function over trajectory distributions, generative policies can generalize effectively to unseen state spaces, offering theoretical guarantees on convergence rates and sample efficiency without assuming a fixed preference oracle.

4.2.2 Sequence Modeling and Transformer-Based Approaches

Recent advancements have reframed reinforcement learning as a conditional sequence modeling problem, leveraging the powerful contextual reasoning capabilities of Transformer architectures. Unlike traditional temporal-difference methods that rely on bootstrapping, these approaches treat trajectories as sequences of states, actions,

and returns, enabling the direct generation of optimal actions via autoregressive decoding. By conditioning the policy on desired return-to-go values, models such as the Decision Transformer effectively synthesize high-performing behaviors from sub-optimal offline datasets without explicit value function estimation, thereby mitigating the distributional shift issues prevalent in conservative Q-learning algorithms [14].

The architectural flexibility of Transformers allows for the integration of diverse modalities and long-range dependencies that recurrent neural networks often struggle to capture due to vanishing gradients. In this paradigm, the attention mechanism facilitates the dynamic weighting of historical interactions, ensuring that critical decision points within a trajectory exert appropriate influence on current action selection. Furthermore, bi-directional encoding strategies have been adopted to reconstruct latent representations by leveraging both past and future context within offline episodes, significantly enhancing the fidelity of hindsight information extraction and improving sample efficiency in data-scarce regimes.

Despite their promise, these sequence-based methods face challenges regarding computational complexity and sensitivity to hyperparameters such as context length and embedding dimensions. Empirical studies suggest that while increasing sequence length provides richer temporal context, it does not always yield proportional performance gains and can impair training efficiency. Consequently, current research focuses on optimizing architectural components, such as reducing the number of attention heads or employing hybrid encoder-decoder structures, to balance model capacity with practical deployment constraints while maintaining robustness across diverse robotic manipulation and locomotion tasks.

4.2.3 Inverse Bellman Operators and Direct Preference Losses

Inverse Bellman operators provide a mechanism to recover reward structures or value functions directly from preference data without explicitly modeling the transition dynamics. By inverting the standard Bellman backup, these methods formulate the learning objective as a fixed-point problem where the optimal value function must satisfy consistency constraints derived from human or oracle preferences. This approach effectively bypasses the need for iterative environment in-

teraction, allowing algorithms to leverage offline datasets containing trajectory comparisons. The theoretical foundation relies on establishing that the preference model induces a unique solution to the inverse Bellman equation, ensuring that the learned value function aligns with the underlying latent reward signal [11].

Direct preference losses complement this framework by optimizing policy parameters or reward models through pairwise comparison objectives rather than cumulative return maximization. Instead of regressing to scalar reward values, these losses minimize the discrepancy between predicted preference probabilities and observed labels, often utilizing Bradley-Terry or Plackett-Luce models. Recent advancements integrate these losses directly into the actor-critic update rules, where the critic is trained to distinguish preferred actions from non-preferred ones via a contrastive objective. This formulation enhances stability by avoiding the propagation of errors inherent in multi-step return estimation, as the gradient signal originates directly from the relative ranking of actions within the dataset.

The synergy between inverse Bellman operators and direct preference losses enables robust offline reinforcement learning from human feedback. While inverse operators ensure global consistency of the value function across the state space, direct losses provide fine-grained discrimination at the action level, preventing overestimation biases common in standard Q-learning. By combining these techniques, algorithms can achieve sample efficiency comparable to online methods while maintaining the safety guarantees of offline training. Furthermore, regularization terms derived from divergence constraints are often incorporated to prevent the policy from deviating too far from the behavior distribution, ensuring that the learned preferences generalize well to unseen states without compromising operational safety.

4.3 Hybrid Offline-Online Fine-Tuning Paradigms

4.3.1 Pre-training on Static Datasets with Online Adaptation

This paradigm leverages large-scale static datasets to initialize policies or reward models, effectively warm-starting the learning process before engaging in online interaction. By pre-training on diverse, offline collections of human demonstrations

or preference comparisons, agents acquire robust representations and baseline behaviors that mitigate the sample inefficiency and safety risks associated with cold-start online reinforcement learning. This initial phase often employs conservative algorithms, such as pessimistic value estimation or behavior-constrained optimization, to prevent overestimation errors inherent in out-of-distribution actions while maximizing the utility of prior data.

The subsequent online adaptation phase focuses on refining these pre-trained models through active feedback loops, addressing the distributional shift between static data and dynamic real-world environments. Rather than discarding offline knowledge, hybrid approaches integrate it as a regularizer or a prior, allowing the agent to explore novel states while maintaining stability. Techniques such as adaptive behavior cloning terms or counterfactual prediction modules are utilized to balance exploration with the mitigation of bias, ensuring that the policy evolves beyond the limitations of the original dataset without succumbing to catastrophic forgetting or unsafe deviations.

Empirical evidence suggests that this two-stage strategy significantly accelerates convergence and enhances final performance compared to purely online or offline methods. The pre-trained representations enable faster reward inference and facilitate generalization to tasks not explicitly covered in the static corpus. Furthermore, by grounding online updates in a rich offline foundation, agents demonstrate improved robustness against noisy human feedback and varying data qualities, ultimately achieving superior sample efficiency in complex, high-dimensional domains where direct online experimentation is prohibitively costly.

4.3.2 Dynamics-Aware Correction and Reset-Free Learning

Dynamics-aware correction mechanisms address the instability inherent in learning from noisy human feedback by augmenting initial learning dynamics with a learned correction term. This approach modifies the standard update rule to prevent the system from converging to suboptimal basins of attraction when human interventions are inconsistent or unbounded. By carefully tuning the relative weights between the original dynamics and the correction component, the learner maintains stability while adapting to trajectory-wise feedback, which is intrinsically more challenging

than stepwise signals in offline preference-based settings.

To mitigate the scarcity of labeled data and the high cost of environment resets, recent frameworks integrate reset-free learning strategies with offline reinforcement learning paradigms. These methods leverage fixed datasets of expert trajectories to pre-train policies, utilizing techniques such as experience relabeling and unsupervised pre-training to maximize sample efficiency. By employing pessimistic value estimation and in-sample regularization, algorithms can effectively constrain policy updates to the support of the training distribution, thereby reducing extrapolation errors without requiring extensive online interaction or manual system resets.

The synthesis of explicit dynamics models with conservative offline learning enables robust policy deployment in real-world robotic manipulation tasks. Instead of relying solely on static datasets, these approaches often combine offline representation learning with online force feedback, allowing agents to generate smooth motions even under unknown dynamic parameters. This hybrid workflow demonstrates superior performance in dynamic environments compared to pure behavior cloning, offering a practical solution for deploying safe and adaptable policies on physical hardware where continuous online tuning is prohibitive.

4.3.3 Safe Exploration and Risk-Sensitive Fine-Tuning

Safe exploration in offline settings addresses the critical challenge of distributional shift without requiring hazardous online interactions. Traditional methods often rely on accurate dynamics models or extensive environment sampling, which are impractical for safety-critical domains like robotics and healthcare. To mitigate this, contemporary approaches employ policy-constraint mechanisms, such as behavior cloning regularization, to limit the learned policy within the support of the offline dataset. This strategy effectively reduces exploration error by preventing the agent from executing out-of-distribution actions that could lead to catastrophic failures or unstable learning dynamics during the initial fine-tuning phases.

Risk-sensitive fine-tuning further enhances robustness by integrating pessimistic principles into the reward optimization process. Instead of maximizing expected returns based on a single point-estimate reward model, these algorithms construct

confidence sets around the reward function and solve for policies that maximize worst-case performance. This distributionally robust optimization ensures that the resulting policy remains effective even under the least favorable reward scenarios within the plausible set. By explicitly accounting for uncertainty in human preference data and model predictions, such methods provide theoretical guarantees against over-optimization on noisy or sparse feedback, thereby stabilizing the alignment process.

Advanced frameworks combine these safety constraints with active uncertainty reduction to balance exploitation and cautious exploration. Algorithms often constrain online fine-tuning to a high-confidence subset of the policy space while actively selecting queries that minimize epistemic uncertainty in poorly covered regions. This dual objective allows the system to refine its behavior using minimal additional data while strictly adhering to safety boundaries defined by the offline distribution. Consequently, these techniques enable the deployment of proficient policies in complex, multi-stage tasks where standard trial-and-error approaches would be unacceptable due to the high cost of failure or the inability to reset the environment efficiently.

5 Future Directions

Despite the significant progress in hybridizing offline and preference-based reinforcement learning for robotics, several critical limitations persist. Current frameworks often struggle with the severe distributional shifts that occur when policies trained on static datasets encounter dynamic, unstructured real-world environments. The integration of heterogeneous feedback modalities remains brittle; while methods exist to combine explicit demonstrations with implicit physiological markers, robust fusion mechanisms that can handle noisy, conflicting, or sparse human signals without degrading performance are largely undeveloped. Furthermore, existing models of human bounded rationality frequently rely on simplified assumptions that fail to capture the full spectrum of cognitive biases and fatigue over long-horizon tasks. This leads to reward misspecification, where agents misinterpret satisficing behaviors as optimal intent. Finally, the sample inefficiency of active query selection in high-dimensional state spaces continues to impose a prohibitive burden on

human operators, limiting the scalability of these systems beyond controlled laboratory settings.

Future research must prioritize the development of unified architectures capable of seamless multi-modal fusion and adaptive personalization. A primary direction involves creating meta-learning frameworks that dynamically weight different feedback sources—such as neural indicators, physical corrections, and pairwise comparisons—based on real-time confidence estimates and context awareness. This would allow agents to rely more heavily on implicit signals during high-cognitive-load scenarios while soliciting explicit corrections only when uncertainty peaks. Additionally, there is a pressing need to advance causal modeling of human intent to distinguish between genuine preferences and artifacts of human error or fatigue. By incorporating causal inference into reward learning, systems can better generalize across diverse users and tasks, moving beyond static user models to evolving representations that adapt to individual behavioral trajectories over time. Research should also explore hierarchical curriculum strategies that automatically generate training curricula based on the agent’s current competence and the user’s demonstrated patience, thereby optimizing the trade-off between exploration safety and learning speed.

The successful realization of these future directions holds the potential to fundamentally transform robotic deployment in complex, human-centric environments. By overcoming the barriers of sample inefficiency and reward ambiguity, next-generation systems will be able to learn safe, personalized policies with minimal human intervention, making advanced robotics accessible for non-expert users in domains such as healthcare, elder care, and disaster response. Enhanced robustness to distributional shifts will enable agents to operate reliably in unpredictable settings, reducing the risk of catastrophic failures during real-world execution. Ultimately, bridging the gap between offline priors and iterative preference refinement will foster a new era of collaborative intelligence, where robots act not merely as tools executing pre-programmed instructions, but as adaptive partners capable of understanding and aligning with nuanced human values and intentions.

6 Conclusion

This survey has systematically examined the convergence of offline and preference-based reinforcement learning as a critical pathway for deploying robotic systems in unstructured environments. We explored how hybridizing pre-collected datasets with iterative human feedback mitigates the risks of online exploration and the ambiguities of sparse reward specification. Key findings highlight the efficacy of synthesizing explicit demonstrations with preference corrections to establish robust behavioral priors while resolving suboptimal segments through fine-grained comparisons. Furthermore, the integration of implicit physiological markers and the modeling of bounded rationality allow agents to interpret noisy, satisficing human behaviors as probabilistic expressions of intent rather than errors. By employing adaptive interaction frameworks, including curriculum learning and active query selection, these methods significantly enhance sample efficiency and enable dynamic personalization that evolves with user needs.

The significance of this work lies in its unique synthesis of disparate methodologies into a cohesive taxonomy that addresses the specific challenges of real-world robotic deployment. Unlike previous reviews that treat offline learning and preference optimization in isolation, this survey elucidates how their intersection overcomes the brittleness of pure imitation learning and the high query complexity of standard preference-based approaches. By critically assessing techniques for fusing multi-modal signals and handling human irrationality, we provide a structured foundation for understanding how heterogeneous feedback can be transformed into dense, informative reward landscapes. This comprehensive analysis clarifies the relationships between current methods, offering researchers a clear perspective on how to balance safety, efficiency, and alignment in complex, dynamic settings where traditional reinforcement learning fails.

Looking forward, the field must prioritize the development of scalable frameworks capable of handling the full spectrum of human rationality and environmental stochasticity. Future research should focus on refining models that seamlessly integrate implicit neural feedback with explicit corrections to reduce the cognitive load on human operators while maintaining high fidelity in reward

shaping. There is an urgent need for standardized benchmarks that evaluate these hybrid systems across diverse physical tasks, ensuring that theoretical advancements translate into robust, deployable solutions. Ultimately, realizing the vision of adaptable and personalized robotic agents requires a sustained collaborative effort to bridge the gap between algorithmic efficiency and nuanced human intent. By addressing these open problems, the community can accelerate the transition from controlled laboratory experiments to safe, effective human-robot collaboration in the real world.

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